AI WORKPLACE READINESS

VerifAI

Measuring the judgement behind AI use - not the tool knowledge - so employers can trust a graduate’s readiness before day one.

22 OCTOBER 2026 ITMDA3-34 EDUVOS BEDFORDVIEW 5 SPEAKERS

FIVE SPEAKERS, FIVE SECTIONS

Who presents what

#	SPEAKER	SECTION	SLIDES	TIME
1	**Meshaque Kalonji**	Introduction - the problem, the product, the promise	3–5	3:00
2	**Vusani Mulaudzi**	The experience - five hubs and a live walkthrough	6–8	3:00
3	**Theodore Nangara**	The assessment engine - formats, scoring, feedback	9–11	2:30
4	**Lwandle Mvelase**	Readiness, motivation and the learning ecosystem	12–14	3:00
5	**Mpho Mokoena**	Engineering, quality, plan versus delivery	15–19	3:30
-	**Meshaque**	Conclusion	20	1:00

Planned

The sprint scope, straight from the project board.

Built

A live, tested product across five hubs.

Left

Testing phase, two content gaps, hardening.

34.5k
LINES OF CODE

107
API ENDPOINTS

241
AUTOMATED TESTS

144
SCENARIO TASKS

THE PROBLEM

AI knowledge is easy to test. Judgement is not.

Employers are adopting AI faster than graduates can be assessed on using it **well**. Tool knowledge is easy to quiz and easy to forget. What actually decides whether a junior hire is trustworthy with AI is the judgement around it.

Catch what is wrong

A hallucinated citation. A fabricated figure. Code that silently swallows an exception. The output looks confident and is wrong.

Know when not to automate

Some decisions carry legal, financial or human consequences. The correct answer is sometimes “use AI with a human in the loop” - or not at all.

Spot the risk first

Privacy, bias, POPIA and client confidentiality. By the time a risk reaches a client, the damage is already done.

> No CV states it. No multiple-choice quiz measures it. That gap is the product.

THE PRODUCT

An app that measures AI workplace readiness

VerifAI puts the learner inside realistic, employer-style work: a scenario containing a flawed AI output or a judgement call, where they must find the problem, fix it, and justify the decision. Every graded attempt feeds a live readiness profile - the number a recruiter could actually trust.

01
Learn
Short AI-literacy lessons and capstones

→

02
Practise
Realistic flawed-output scenarios

→

03
Be assessed
Server-scored, exam-style conditions

→

04
See evidence
Readiness, tier and full attempt history

→

05
Keep going
Streaks, XP, tokens, rewards

Finance

A gross-profit summary with a payment that never arrived.

Legal

A case summary citing a judgment that does not exist.

IT / Software

AI-written code that swallows an exception silently.

Business Consulting

A recommendation that ignores a client's POPIA duties.

WHAT A RECRUITER SEES A 0–100 readiness score, a tier, and an evidence trail of every scenario attempted - not a certificate for watching videos.

THE CORE LOOP

One action, then a number that moves

Home never opens on a wall of charts. It opens on the one thing worth doing next. Submitting a task updates readiness, XP, streak and achievements **in a single transaction** - the server is the source of truth, and the app only renders what it is told.

01 **Home recommends the next task**
Unseen first, in a difficulty band suited to level; least-recently attempted rotates back in as “Practise again”.

02 **The learner answers**
Pick the flawed lines and write the fix, choose the correct option, or make a decision plus a justification.

03 **The server scores and commits**
Attempt, analytics row, coverage, readiness, XP, tokens, streak and achievements - one transaction.

04 **Feedback explains, then offers the next step**
What was right, what was missed, and a full-screen next task.

05 **Home reflects the new reality**
Readiness, tier, XP and a fresh recommendation.

One request, not five

Home is `GET /api/dashboard`: profile, progress, dimension scores and one recommended task. It replaced five calls, three of which ran an expensive selection query and were then discarded.

Practice is honest

Repeat attempts are labelled **Practise again** and earn no new coverage - a practice pass pays half XP, once. You cannot farm readiness by re-doing the same task.

Server-side “today”

Streaks are anchored to **Africa/Johannesburg**, not the phone’s clock or the server’s UTC - a check-in just after midnight cannot land on yesterday.

NAVIGATION

Five hubs. Everything one tap away.

The shell is deliberately shallow: five areas on a floating dock, with every other screen pushed from the hub that owns it. A full index drawer stays available as a secondary route for people who prefer a list.

Home

Recommended task, readiness and tier, XP bar, streak and completed tiles. Pull to refresh.

Learn

AI Lab lessons → chapters → topics with sources; recall practice; capstones; roadmap; Ask Vera.

Practice

Verification, AI Judgement, Risk & Governance, Verification Studio, interview bank, industry cases, history.

Progress

Readiness breakdown, XP/level, activity chart, leaderboard, milestones, history, rewards.

Profile

Identity, track change, photo, appearance (light/dark/system), wellness centre, Vera, staff admin portal, sign out, delete account.

Built for real use

Material 3, bundled Inter/Manrope, light and dark themes, respects OS text scale, semantic labels on custom-drawn scores and cards, inline retry on a dropped connection.

40
SCREENS

43
ROUTES

5
DOCK TABS

3
THEMES

WHAT IS ACTUALLY BUILT

The screens behind each hub

HUB	DESTINATIONS - ALL IMPLEMENTED AND NAVIGABLE
Home	Dashboard with next task, readiness ring, XP, streak, completed count, dimension index; drawer index; task runner; feedback
Learn	Lab catalogue → lesson → chapter → topic; recall practice; capstone list → exam paper → result; skill roadmap; Ask Vera
Practice	Verification card set → full-screen runner → feedback; AI Judgement; Risk & Governance; Verification Studio (paste / upload / link / citation); interview bank; industry cases; task history
Progress	Readiness breakdown by dimension; XP and level; 30-day activity chart; leaderboard (global + track); milestones; paginated history with deep-linked feedback; rewards catalogue and redemptions
Profile	Account and track; photo; appearance; wellness (mood, journal, self-care, breathing, weekly summary); Vera sessions and memories; staff admin entry; delete account
Auth / first run	Splash; welcome tour (5 steps); sign in, register, reset; industry selection; track change

Full-screen task runner

Verification tasks run full screen with the learning assistant in the app bar - not a bottom sheet. Continue opens the next task the same way.

Deep links survive

`/feedback/:attemptId` re-fetches the attempt, so a restart or a shared link still lands on real feedback.

Staff separated

Admin entry appears only for accounts with a database role (`owner` / `moderator`) - never a client-side flag.

22
FEATURE MODULES

43
ROUTES

5
HUBS

3
THEMES

LIVE WALKTHROUGH

Eight steps, about four minutes

This is the script for the live demo. If the network or a free-tier cold start gets in the way, fall back to the recorded clip and continue from step 5.

- **1. Sign in** - existing learner, Finance track, so Progress already has evidence.
- **2. Home** - point at readiness, tier, streak and the recommended task.
- **3. Practice → Verification** - open a line-text task full screen.
- **4. Answer badly, on purpose** - tap one wrong line, skip the fix. Show the honest feedback: the reason alone cannot pass.
- **5. Retry and pass** - correct lines, real fix, real reason. Feedback shows the 30 / 40 / 30 split.
- **6. Continue** - next task opens full screen, no friction.
- **7. Progress** - readiness coverage stepped up; open history and a deep link.
- **8. Learn → AI Lab** - open a capstone paper. Then Profile → Wellness for the second half of the product.

144	576	4	4
TASKS SEEDED	ANSWER OPTIONS	CAPSTONES	INDUSTRY TRACKS

THREE LIVE SKILL ENGINES

Different skills need different answer formats

One shared scoring pipeline, three genuinely different task shapes. Nothing is graded by an LLM on the critical path - the score is deterministic, auditable and cheap.

SKILL	FORMAT	SCORING	CONTENT
Verification	**Choice** - four options, one correct. **Line-text** - numbered AI output; tap the flawed lines, then type the fix and the reason.	Choice: 100 / 0, pass ≥ 60. Line-text: flaw **30** + fix **40** + reason **30**, proportional criteria matching with variants, prefix and numeric-compact matching.	60 24 choice 36 line-text
AI Judgement	Scenario → **decision** (use AI / use with human review / don't use) + best-of-three **reason**. Optional rationale.	Decision **50 / 25 / 0** (correct / neighbour / opposite); reason **50 / 0**. Pass ≥ 50.	60 15 per track
Risk & Governance	Scenario → **decision** (proceed / don't proceed) + best-of-three **reason**. Optional rationale.	Decision **50 / 0**; reason **50 / 0**. Pass ≥ 50.	24 6 per track
Interview	Per-track question bank with model answers and what employers are testing.	Ungraded. Marking practised feeds **10%** of readiness as coverage. Idempotent.	14

WORKED EXAMPLE A numbered AI summary - "Revenue grew 15% because 2023 outperformed" - flag **line 2**, fix it to **"compare like periods"**, reason **"the prior quarter covered 14 weeks"** → **30 + 40 + 30 = 100**. Prompt Design is scaffolded in the model but has no content yet.

FAIRNESS RULES

Scoring that cannot be gamed

Verification line-text: the fix is the skill

At a flat 60, tapping the flawed lines (30) plus a plausible reason (30) passed without fixing anything. Line-text therefore requires **≥ 70 and** a non-zero flaw **and** fix score. The reason alone can never carry the task.

XP is earned, never printed

XP is score-proportional with a difficulty ceiling - easy 50, medium 70, hard 100. A wrong answer earns **0 XP**. Practice pays half, once.

Options never leak their score

The task payload omits every option's score and the correct flag. The answer key exists only in the database until the attempt is scored.

One transaction, six effects

A submission commits atomically:

- the attempt **and** its analytics log row
- feature coverage (passed ÷ available)
- readiness and tier, only if coverage moved
- XP, level and token grant
- streak advance
- new achievements, token grants batched

Idempotent by construction

The token ledger has a unique `(user, reason, ref)` key, so a retried request cannot double-credit. Practice passes are counted once.

CLOSING THE LOOP

Feedback that survives the session

A score without an explanation teaches nothing. Every attempt is stored with a title snapshot, the selected lines, the rationale and the pass flag - so feedback is a real record, not a route-local variable.

Three honest states

Correct, incorrect and error are distinct screens with different copy and actions. Network failure shows a retry, never a fabricated score.

Re-fetchable

`GET /api/history/attempts/:id` is owner-scoped; `/feedback/:attemptId` renders the stored result after a restart or deep link.

Searchable history

Paginated with limit/offset, free-text search and type filters, plus infinite scroll. Rows open their feedback.

What the learner sees

- Score and pass/fail against the dimension threshold
- The correct lines and a model correction
- Why their reason did or did not land
- XP earned, streak, and any achievement unlocked
- One clear next action: retry, or continue

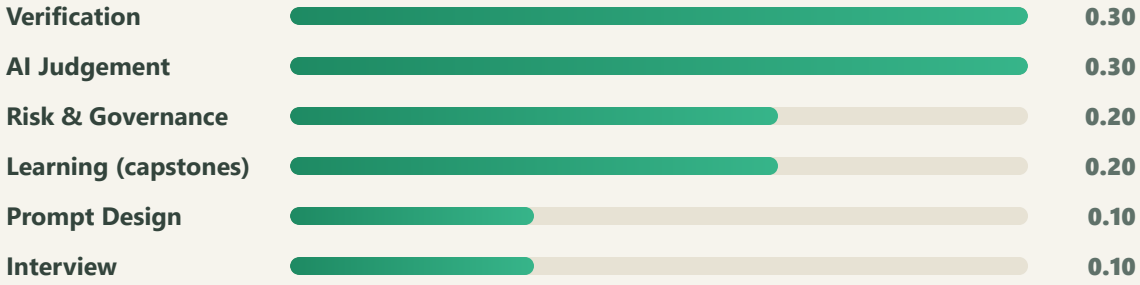
What the team sees

- Answer logs per task and per option
- Per-track progress and funnel in the admin portal
- CSV exports for users, attempts and track progress
- Moderator queue for extended and rubric answers

THE READINESS MODEL

One score, six weighted components

Readiness is a 0–100 weighted mean. The denominator is the sum of the weights of the components the learner's **track actually offers** - so an empty module cannot silently cap a learner, but a module left untouched still counts as zero coverage.



Each skill score is **coverage**: passed tasks ÷ tasks available. A skill with four tasks fills **25 → 50 → 75 → 100%** as each is passed. It climbs with evidence; it does not decay between sessions.

Inputs, not opinions

Skill scores come from stored coverage; Interview from questions practised ÷ questions offered; Learning from the average best capstone result. Nothing is self-reported.

Tiers follow the score

Novice	0 – 49
Developing	50 – 69
Proficient	70 – 84
Advanced	85 – 100

Why it is hard to fake

Readiness is written on the server on every scored attempt, from stored coverage. Three strong answers cannot reach Advanced if two modules are untouched - **coverage counts**.

THE HABIT LOOP

XP for effort, tokens for progress

Readiness measures competence; the reward economy keeps people coming back. XP is never spent. Tokens are - granted by levels and achievements, written to an append-only ledger.

XP & levels

Score-proportional per difficulty, **0 for wrong**. Levels step by +60 XP and unlock spendable tokens.

Streak

Daily, anchored to Africa/Johannesburg. Server dates mean a timezone cannot break it.

Achievements

13 definitions, granted in one batched statement on the submission that earns them.

Rewards

9 catalogue items with original artwork, atomic redemption, and a ledger that cannot double-credit.

Leaderboard & milestones

Global and per-track boards with the learner's own rank always live; the board itself is cached per (track, limit) for 60 seconds. A covering index on `(xp DESC, readiness_score DESC, user_id)` keeps it off a full scan.

Activity, not a number

A shared activity view projects task attempts, lab practice and capstones into one timeline, and the 30-day chart groups on stored dates instead of converting a timezone per row.

THE LEARNING ECOSYSTEM

More than a quiz app

Scenarios test judgement. The rest of the product builds the knowledge and habit around them - and every part feeds the same profile.

AI Lab

5 lessons, 248 sections, 240 activities. Track-specific and global content, per-section progress, a source library with licence metadata, recall practice.

Capstones

4 assessments, 100 questions. Timed, attempt-capped, pass mark and reward. Objective questions auto-mark; extended answers queue for a moderator, with AI suggesting rubric marks.

Verification Studio

Paste an AI answer, upload a PDF/DOCX, drop a link or a citation and get a structured verification report. The submitted material is never persisted.

Industry cases

16 longer workplace briefs, served per track, deliberately unscored - they build the context the scenarios test.

Wellness Centre

Daily mood check-in, private journal with undo, self-care activities, breathing exercise, weekly summary. Private to the learner.

Vera & assistant

An app-wide AI coach and an in-task assistant. Both stream over SSE, keep inspectable memory, and never hand over the answer.

ARCHITECTURE

One API, one database, one source of truth

CLIENT	**Flutter** (Android, iOS, web) - Riverpod state, go_router navigation, Dio HTTP. Screens read a provider; providers call one `ApiClient`. That single seam is what makes 55 widget and flow tests possible without a network.
TRANSPORT	HTTPS with a **Firebase ID token** on every protected call. A 401 interceptor refreshes once, retries, then signs out cleanly.
API	**Node.js 22 + Express** on Render - **20 routers, 107 endpoints**, rate limits, in-process caches, ETags, SSE streaming, a bounded document-parsing worker pool.
DATA	**PostgreSQL on Supabase** - **50 tables**, 29 ledger-applied migrations, a shared pg pool, parameterised SQL throughout.
SERVICES	**Firebase Auth** for credentials and token verification; **NVIDIA-hosted LLMs** for analysis, chat and marking with a fallback model; **Tavily** for live web search.
DELIVERY	**Render** blueprint (migrate, then start) · **Firebase Hosting** for the web build · **GitHub Actions** for tests and the APK artefact.

12.5k
BACKEND JS LINES

21.9k
FLUTTER DART LINES

163
COMMITS

5
TEAM MEMBERS

ENGINEERING DECISIONS

What we fixed after auditing ourselves

Three review rounds, 24 findings closed. The ones that matter most are below - each replaced a working-but-wrong design, not a missing feature.

Security

Every protected route verifies a Firebase token; roles come from `admin_roles` in the database, so a moderator cannot promote themselves. Bodies are size-capped per route, SQL is parameterised, the answer key never ships early, submitted verification material is never stored, and CI runs gitleaks.

Correctness

The readiness denominator is fixed per track, so starting a second module no longer *lowers* the score. Streaks use one configured timezone. Practice passes count half, once. Readiness is skipped when nothing moved.

Testability

One `ApiClient` seam, a mock client for widget tests, and a 200% text-scale regression test so accessibility cannot silently break layout.

Performance

Home is one composed request instead of five; analytics, leaderboard and static catalogues cache for 60 s with strong `ETag`/304; indexes follow the actual queries; the timezone conversion is gone from chart queries.

Reliability

Rate limits live in PostgreSQL, so two API instances share one per-account budget. Document parsing runs off the request thread with byte, page, queue and timeout caps. Streaming aborts upstream work when the client disconnects and sends keep-alives during model silence.

Accessibility

Light, dark and system themes; bundled Inter/Manrope (no runtime font download, no reflow on first paint); OS text scaling; semantic labels for scores, streaks and cards.

QUALITY GATES

Green before it ships

Nothing merges on a red build. CI runs the backend suite, the Flutter suite, static analysis, a dependency audit, a secret scan and an APK build - on every push and every pull request.

186	55	0	0
BACKEND TESTS	FLUTTER TESTS	ANALYZER ISSUES	ANALYZER WARNINGS

What the 186 backend tests cover

- Routing and auth
- Authorisation and roles
- Scoring and thresholds
- Readiness and tiers
- Capstones
- Wellness
- Rewards and tokens
- Lab
- Admin
- Streaming
- Streaks
- Wiring

What the 55 Flutter tests cover

- Auth interceptor
- Login and onboarding
- Task submission
- Lab navigation
- Judgement flow
- Wellness
- Navigation shell
- Feedback states
- Assistant
- 200% text scale
- API error mapping
- Markdown rendering

CI: backend test + `npm audit` · Flutter format, analyze, test · gitleaks secret scan · release APK uploaded as an artefact (14-day retention).

SCOPE, FROM THE PROJECT BOARD

Planned versus delivered

BOARD CARD	WHAT WAS SCOPED	STATUS
Backend - Fix	No points for wrong answers; difficulty-scaled XP; hardened verification scoring; leaderboard; activity history; achievements and milestones; streak timezone.	DELIVERED - 186 tests
DB - Restructure & Reseed	Per-feature tables, shared option table, extended attempts, migration, rewritten schema/seed.	DELIVERED - 50 tables, 29 migrations, 144 tasks
Backend - Build	AI Judgement, Risk & Governance, Interview Practice on a shared choice engine.	DELIVERED - 60 / 24 / 14 items
Frontend - Build	Tasks hub, judgement, risk, interview, feedback, history, promotion display, models and client.	DELIVERED - 5 hubs, 40 screens
Frontend - Fix	Real feedback states, 401 refresh, profile/industry states, offline and timeout, honest labels.	DELIVERED - analyze clean
Beyond the sprint	Not on the original board.	SHIPPED - Studio, AI Lab, rewards, wellness, Vera, admin portal
Not delivered	Prompt Design content; Workshop mobile UI.	CARRIED - see next slide

Every item marked delivered is live against the deployed API and covered by the automated suite. Nothing above is a mock-up.

HONEST STATUS

Outstanding work

Development phase closes 23 October; testing runs to 30 October; the final presentation is 13 November. These are the gaps we know about, in priority order.

1 · Testing phase DUE 30 OCT

QA plan and test cases, a regression pass across every screen, a device matrix (budget Android, iOS, web), and verification of scoring, readiness, tiers and passport rules.

2 · Prompt Design content

Scaffolded in the schema, weights and feature map, but no tasks authored - task creation rejects it for now. It is excluded from readiness until the track offers it.

3 · Workshop mobile UI

Threads, replies, reactions, reports and moderation are complete on the API and in the admin portal. No learner-facing screen yet.

4 · Integration tests

The 186 backend tests mock PostgreSQL, so SQL regressions rely on review and the schema. An embedded-Postgres suite is the highest-value next test.

5 · Offline mode

A dropped connection shows an inline retry rather than a cached read. No offline queue.

6 · Content workflow

Authoring still goes through the admin portal and seed files - there is no dedicated CMS review workflow.

7 · Database hardening

The Supabase tables are reachable through the API only in practice, but row-level security is not enabled on the public schema. Enabling RLS (the API connects as owner) is a low-risk hardening step before submission.

8 · Final deck and rehearsal

This is the progress review. The 13 November presentation needs the final numbers, a rehearsed demo and Q&A preparation.

WHERE WE ARE

A working product, not a prototype

VerifAI turns a vague hiring question - *"can this graduate be trusted with AI?"* - into something measurable, and backs it with the engineering that makes it credible.

- **The learner** works through realistic, flawed AI output and is scored on the judgement behind the answer.
- **The result** accumulates into a readiness score, a tier and a body of evidence an employer can inspect.
- **The habit** is supported by XP, tokens, streaks, rewards, a wellness centre and an AI coach.
- **The engineering** is a single authoritative scoring pipeline, a schema that evolves through reviewed migrations, and graceful degradation for every AI feature.
- **The delivery** is real: deployed API, deployed web build, installable APK, tested on every push.
- **The next phase** is testing and hardening - not new features.

144	241	107	5
SCENARIO TASKS	AUTOMATED TESTS	API ENDPOINTS	SHIPPED HUBS

DELIVERY CHECKLIST

Things easy to forget tomorrow

Warm the API before you start

The API runs on a free tier that sleeps when idle. Open the web app **5 minutes before** you present so the first request is not a 30–60 s cold start in front of the panel.

Confirm your slot length

This deck assumes ~16 minutes plus questions. If you have 10, cut slides 7 and 13 first - keep the demo and the readiness model.

Demo with a lived-in account

Use a learner who already has attempts, so Progress, history and readiness look alive. A brand-new account makes the product look empty.

One driver, four talkers

Agree who holds the laptop. Handover cues: "…and Theodore will show how that is scored." Rehearse the switch at least twice.

Two fallbacks, not one

Install the APK on a phone **and** keep a 60–90 s screen recording of the full loop. If the network dies, play the clip and keep talking.

Say the same numbers

Use the figures in this deck consistently: 144 tasks, 107 endpoints, 241 tests, 50 tables. Mixed numbers invite doubt.

Have the links ready

Web: verif-ai-5a435.web.app · API health: `/api/health`. Put a QR code on the last slide or a handout.

Be first to name the gaps

Prompt Design content and the Workshop screen are unfinished. Say so before the panel finds it - it reads as control, not failure.

Q&A PREPARATION

The questions you will be asked

QUESTION	SHORT ANSWER
Why not let an LLM grade the answers?	Determinism, fairness and cost. A scored task must give the same result twice and be auditable. LLMs are used where judgement is genuinely open - rubric suggestions for extended answers - with a moderator making the final call.
How do you stop people gaming readiness?	Coverage denominator fixed per track, distinct passed tasks only, server-side scoring, practice passes count once at half value, and the answer key never leaves the database before submission.
What about POPIA and privacy?	Submitted verification material is never persisted; API keys stay server-side; Vera receives only bounded context with explicit privacy guidance; wellness content is private to the learner.
Does it scale / what does AI cost?	60-second caches, ETag/304 on static content, per-account rate limits shared across instances, a fallback model, and deterministic scoring on the critical path so AI is never in the hot loop.
What is still missing?	Prompt Design content, Workshop mobile UI, a real-Postgres integration suite, offline mode, and the testing phase (regression, device matrix) due 30 October.
Is it actually deployed?	Yes: API on Render, database on Supabase, web build on Firebase Hosting, and a release APK built by CI. Every flow demoed runs against the live API.

144
TASKS · 60/60/24

50
TABLES · 29 MIGRATIONS

186 + 55
BACKEND + FLUTTER

4
TRACKS